PRG110
Visual Analytics

Skriftlig gruppeeksamen

Høyskolen Kristiania

Visual Analytics Eksamen

Våren 2023

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Dataset Description:

We have received two datasets, one contains detailed information about Company X and different data around their business. This includes customer-specific information, product details, and financial details. The second dataset is related to the UN's annual World Happiness Report, which ranks countries based on the happiness levels of the inhabitants and several factors that affect their happiness levels. Through both these datasets we try to extract valuable insights for the business and the public that we believe will be beneficial to look at.

By looking at the data types we can see there are two different types of data, categorical data and continuous data.
During a lecture in Visual Analytics, 9 February 2023 our lecturer described categorical data as a form of information that can be stored and identified based on their names or labels. It is a type of qualitative data that can be grouped into categories instead of being measured numerically. With dataset 1 you have delivery mode, segment and city as examples for categorical data while in dataset 2 you have country and regions as other examples of categorical data.
Continuous data describes information that can take virtually any value, this includes things like height, weight or any kind of numerical measurement (iSixSigma 2022). In dataset 2 you have happiness rank- and score as examples while in dataset 1 you have sales, quantity and profit numbers that are continuous (numerical) data.

By analyzing the different data in Tableau, we can achieve several objectives. By using Tableau we can create multiple views and graphs from the data. We can track sales development over time, identify the most profitable products, analyze sales geographies, and conduct a segment analysis of our customers. With the help of the data we can get a better understanding of the customers, how the company can optimize their product selection and ultimately improve profitability.

For example, by analyzing "Profit" against "Product Name", we can identify which products are generating the most profit for us. We can also analyze sales trends over time (using "Order Date") to identify busy periods or seasonal patterns in sales. By examining "Sales" and "Quantity" together, we can gain an understanding of which products are selling in the highest volumes.

For dataset 2 we can use this information to visualize happiness levels globally and investigate which factors most clearly correlate with high happiness. We can also examine changes over time, both within individual countries and across regions. This information can give us insight into effective strategies to improve well-being across different communities and get a better understanding of the different regions and what is important for them and their happiness level.

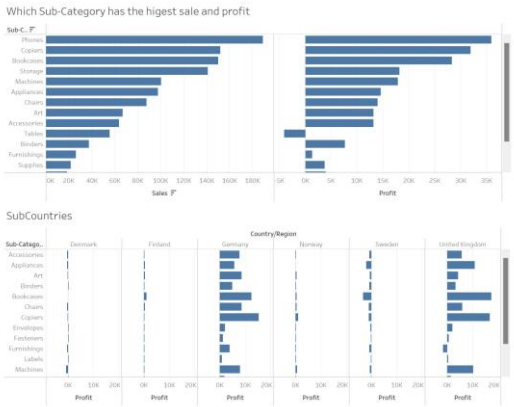
Dashboard: Design

Our main object when it comes to design was to keep it simple, clear and efficient. So that the audience we reach out to, understand the message behind the visualization. As Stephen Few said it so well, ""*most dashboards fail to communicate efficiently and effectively, not because of inadequate technology, but because of poorly designed implementations. No matter how great the technology, a dashboard's success as a medium of communication is a product of design, a result of a display that speaks clearly and immediately*."(Few 2006, 12).

Stephen Few mentions a couple of mistakes that are quite common in dashboard design. These mistakes that are mentioned are exceeding the boundaries of a single screen, displaying excessive detail or precision, choosing inappropriate display media, cluttering the display with useless decoration and misusing or overusing color.

"*One of the great benefits of a dashboard as a medium of communication is the simultaneity of vision that it offers: the ability to see everything that you need at once*."(Few 2006, 39). When preparing the dashboards one of our goals was to avoid spreading the different visualizations over different dashboards. We want the audience to get enough information from the dashboard that is presented to the target audience and avoid any going back and forth between pages so that they understand the message. That is also why we wanted to avoid any form of scrolling. So that the target audience isn't getting the feeling that they are lacking any information.

Another objective for us was to avoid too much clutter and keep it simple. "*Clutter are visual elements that take up space but don't increase understanding. There is a simple reason we should aim to reduce clutter: because it makes our visuals appear more complicated than necessary.* "(Knaflic 2015, 73) We wanted the target audience to avoid getting a cognitive overload and avoid unnecessary and distracting decoration. By for example hiding several fields on the x- and y axes we avoid unnecessary clutter, shown by the examples below. The tooltips were updated so if the audience wanted more information it is available. By hovering with the pointer over the area where information is needed it will show up.



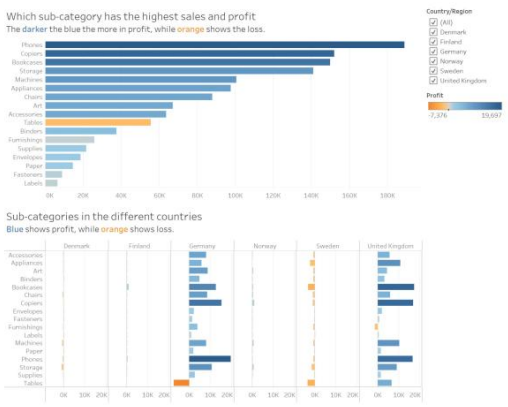


Figure 1 & 2: Figure 1 shows it untouched, while figure 2 is the result of adding color, removing field and assembling two of the graphs into one.

Coloring was also a specific choice by us, knowing that by choosing specific colors we might exclude parts of the audience. As Stephen Few mentioned, color choices should be made

thoughtfully with an understanding of how we perceive color and the significance of color differences. We merrily assume that we can use colors such as red, yellow and green to assign important meanings to data, but in doing so we exclude the 10% of males and 1% of females who are color-blind (Few 2006, 62). Though we might have wanted more variation in our coloring, we resisted the urge to use color for the sake of being colorful; instead, we used the coloring to leverage important parts of our visual. As Knaflic pinpointed, when used sparingly, color is one of the most powerful tools you have for drawing your audience's attention (Knaflic 2015, 117). Examples of ways we used the coloring is by adding color in the subtitles with a short explanation and by having two different colors showing the difference between less and more.

When it comes to the choice of visualization we mostly went for bars and line graphs. Those were the two visualizations we were most familiar with but also felt safe would communicate what insight we had. As Knaflic mentioned, bar charts should be leveraged because they are common, as this means less of a learning curve for your audience. Instead of using their brain power to try to understand how to read the graph, your audience spends it figuring out what information to take away from the visual (Knaflic 2015, 50).
In both datasets we have different regions and categories and bar graphs are great for displaying measures that are associated with items in this kind of groupings (categorical). While the reasoning behind line graphs, they are most commonly used to plot continuous data. Because the points are physically connected via the line, it implies a connection between the points that may not make sense for categorical data (Knaflic 2015, 45). Also Few mentioned that when we want to emphasize patterns in the data, such as trends, fluctuations, cycles, rates of change, and how two data sets vary in relation to one another, line graphs provide the best means (Few 2006, 120).

Dashboard: Development

We began mocking up sample dashboards and tried different variations of the insights we wanted and through several tries and redesigns we got closer to what we think worked.

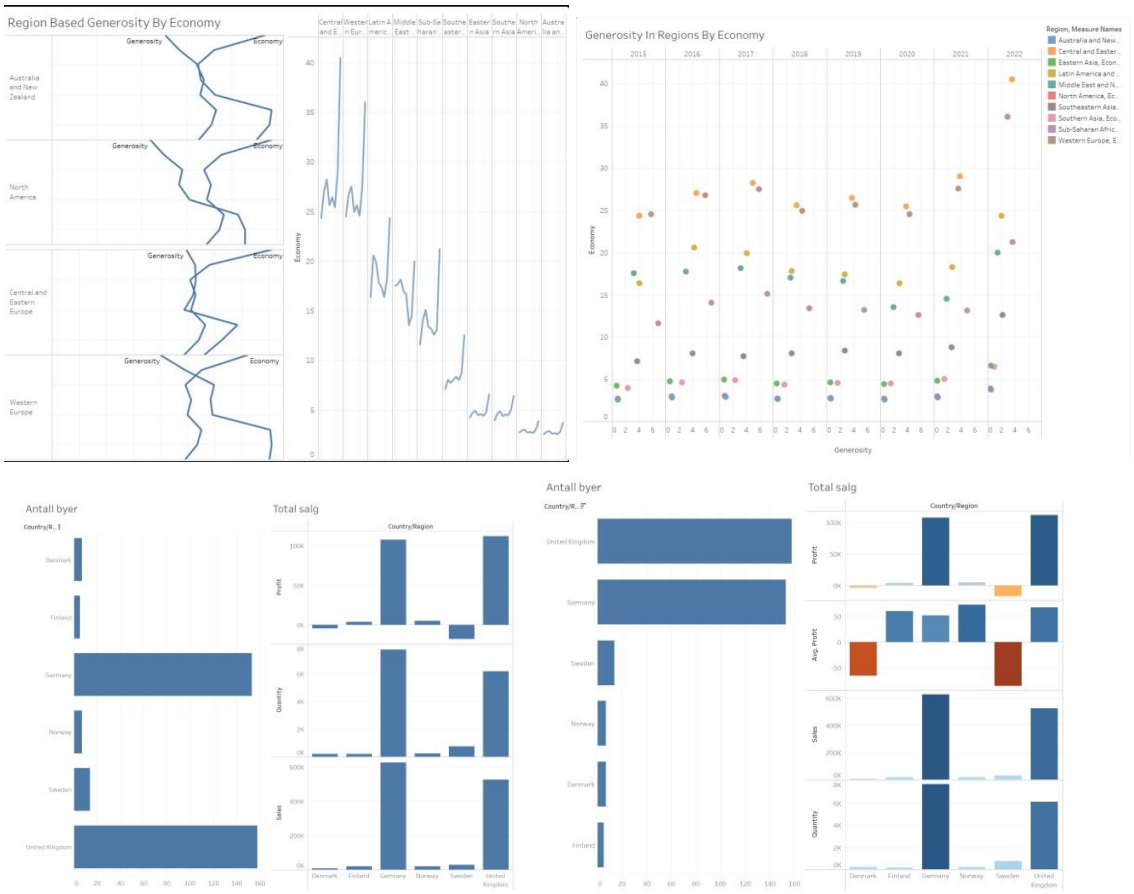


Figure 3,4,5 & 6: Figure 3 and 5 both to the left are first drafts, while 4 & 6 are the final designs.

These are two examples, where the visualization to the left was how they first looked, and after some adjustments how the end result looked.

We threw whatever we had in mind on the wallboard and talked it out. What were the pros and cons with the different visualizations we presented to each other. What did the visualization communicate to us and what did the audience get out of it. We even showed the insights to people that weren't in our group to get a fresh perspective on it. Because one of the pitfalls of getting too attached to our work is that we forget that the audience might not have any knowledge and therefore have a totally different perspective to it all. When it came to the technical part of using Tableau it went well. Most of our knowledge came from completing the tasks given to us through the semester, and we didn't need too much external help. We played around with the tool and learned a lot just by sparring with each other and what we experienced with our experimentation.

<u>Ten meaningful insights:</u>

Dashboard 1:

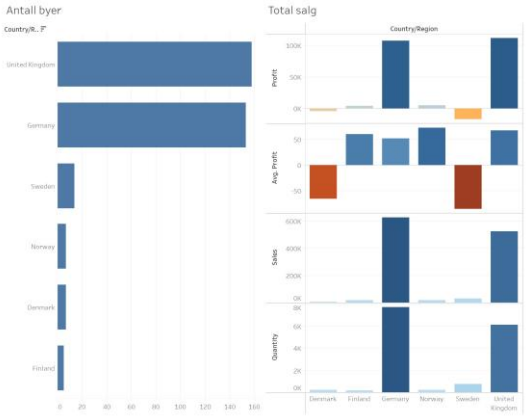


Figure 7: Overview of the amount of cities Company X has gotten orders from and a view of total profit, average profit, sales and quantity.

Insight 1:Company X has orders from several towns but specifically there are two countries that most of the orders origin from. Most of the profit comes from the UK and Germany. But when you take the average in profit the gap gets a lot less and the UK and Germany actually fall down the ranking. Norway has the highest average profit while the UK is second. Fully shows that the company can profit from the minor countries.

Insight 2**:** Sweden is the country where Company X has the largest loss. Even though Sweden is the third country overall when it comes to sales and quantity. Therefore it is recommended to take a deeper look into the Swedish numbers and why they are in so much loss. What kind of products are getting sold is of importance and what are they experiencing.

Dashboard 2:

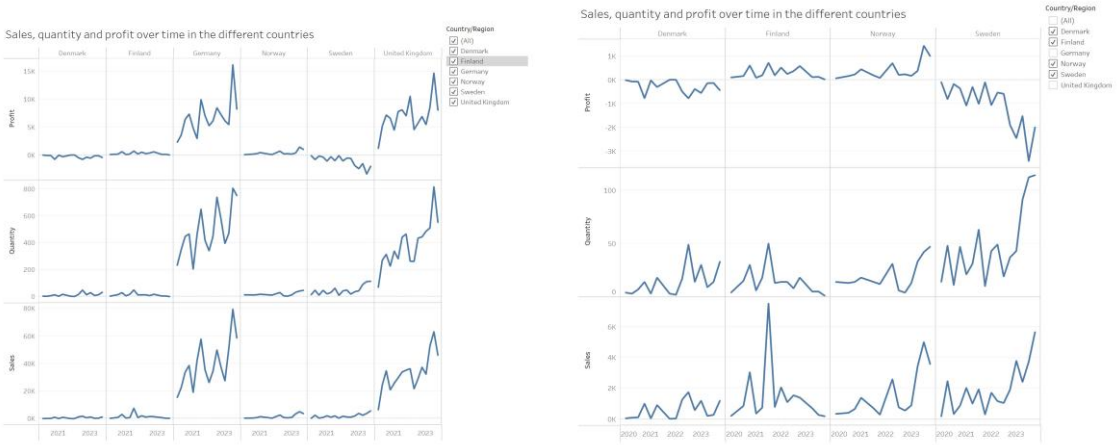


Figure 8 & 9: Figure 8 shows sales, quantity and profit over time for all the countries. While in figure 9 both UK and Germany are excluded.

Insight 3 - During the years there has been a steady increase in revenue, sales and quantity for the UK and Germany. When you filter away the two nations, the same can't be said about the other countries but especially Sweden is a curious case. As mentioned earlier

they have increased their sales and quantity over time but they keep losing money. So it is important to look at what kind of products they are selling so that the other countries don't fall in the same trap. While Norway has had an increase in all three groupings, the other countries can have something to learn from their experiences, specially Sweden and Denmark that are going in minus.

Dashboard 3:

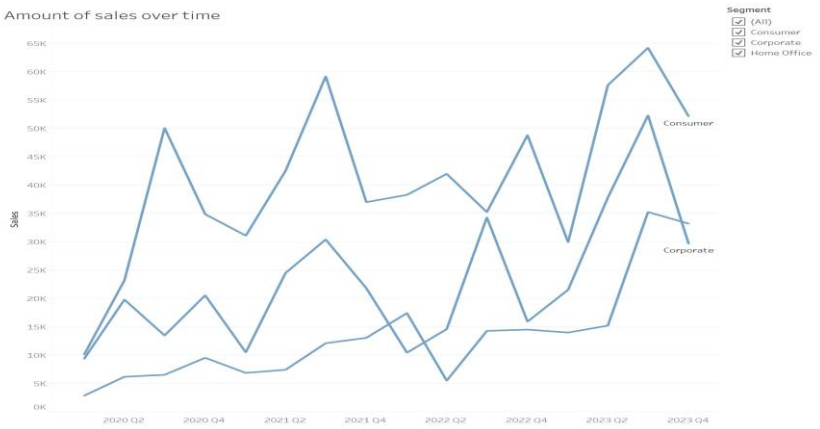


Figure 10: Amount of sales for the different segments over time.

Insight 4 - All the segments have had an increase of sales over time, but all three had a fall in sales during the last quarter. The consumer segment has been the top segment since the start and keeps a safe distance to the other segments through the years. While corporate has been in a secure second place since the start, they were recently overtaken by the home office segment in the last quarter. We might see a new trend or just a one-off case. The fact remains that the consumer segment is still the most important segment for the company and shouldn't be neglected.

Dashboard 4:

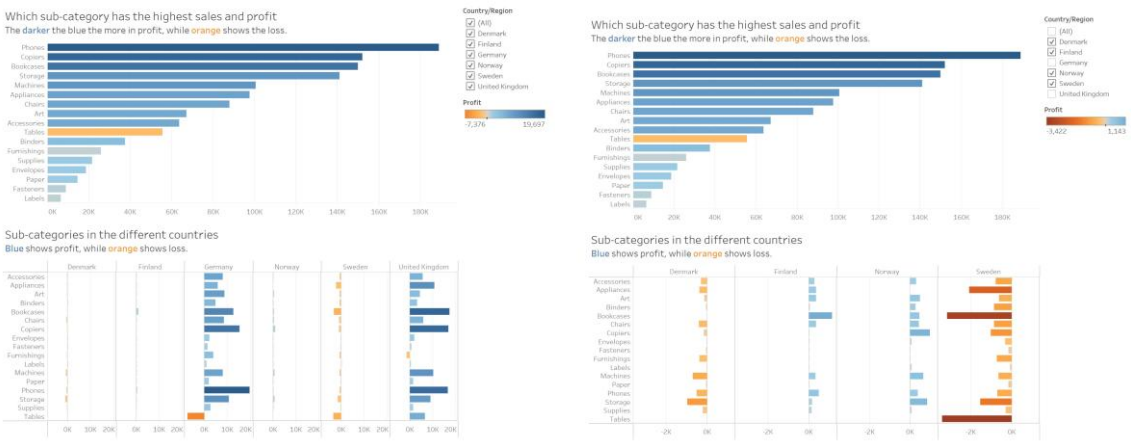


Figure 11 & 12: Figure 11 is an overview of which sub-categories sold and profit most. Also shows the sales & profit in the other countries. Figure 12 shows the same but excluding UK and Germany in the barcharts below.

Insight 5 - Here you can see which subcategories Company X sells the most and also earn the most profit from. Phones, copiers and bookcases are the top three when it comes to sales and profit. On the other side, tables are the one subcategory the company is losing money on. Even though the sales are midtable, Company X is losing money by selling tables. Be aware of having a focus on this subcategory knowingly the company is at loss, but also know which subcategories the company earns a profit on.

Insight 6: By going in depth, you can see that both Sweden and Germany have lost money when selling tables but the UK has earned a profit. Recommend going in dialog with the UK team to find out why they are profiting from the sales while the other two countries ain't. What is the difference maker here between these countries, so that the Germany and Sweden team can turn things around and get a profit.

Dashboard 5:

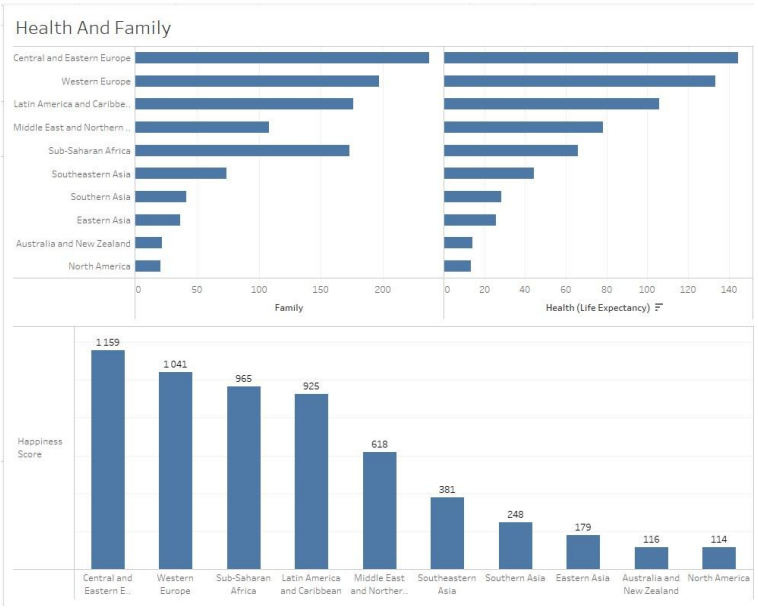


Figure 13: Regional Health/Family

Insight 7: By looking at the data we can see the close correlation in Happiness Score by Family and Health in each region. All regions which score high in Family and Health, also score high in Happiness Score.

Dashboard 6:

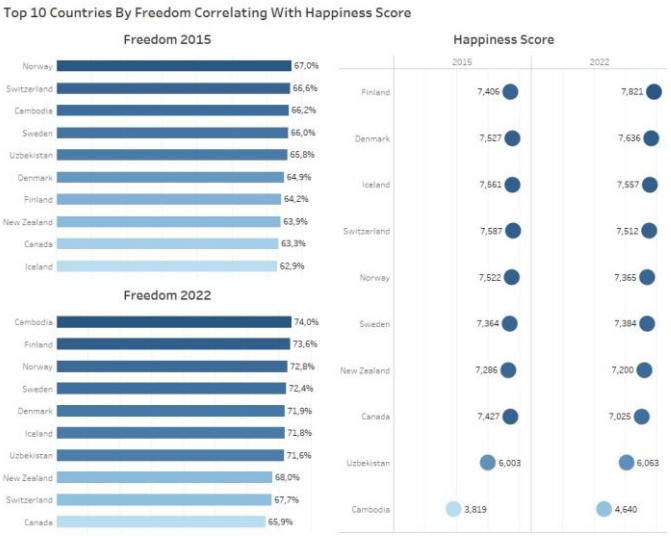


Figure 14: Top 10 Freedom 2015/2022

Insights 8: Countries scored highest in Freedom in 2015 and 2022 who's Freedom have gained significantly, have gained in Happiness Score. Countries with minimal gain in Freedom have either gained minimal points or lowered their points in Happiness Score.

Dashboard 7:

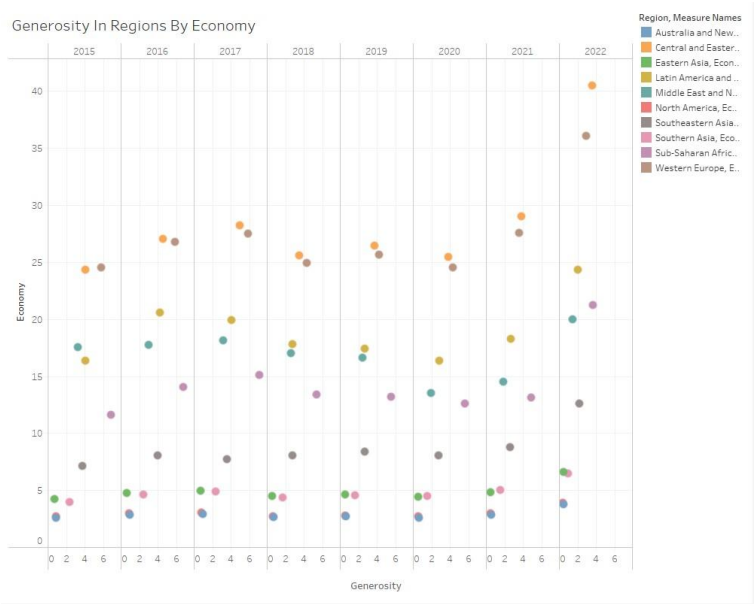


Figure 15: Scatterplot Generosity/Economy

Insight 9: Comparing Generosity by Economy the data shows the correlation by each region usually having higher points in Generosity if their Economy scores high. Standing out, Sub-Saharan Africa scores the highest in Generosity even though their Economy is not the highest.

Dashboard 8:

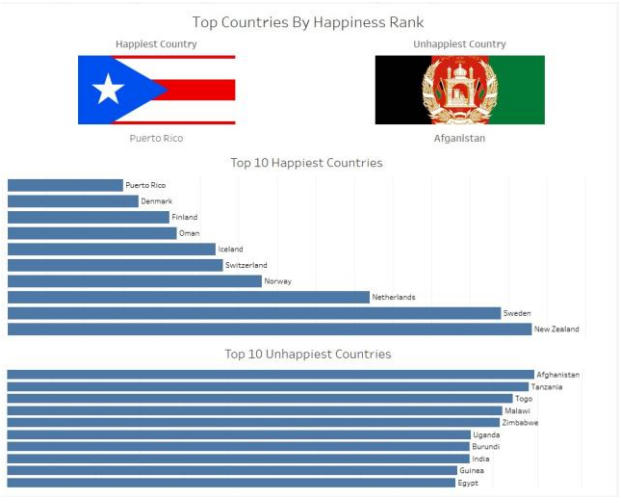


Figure 16: Top 10 Happiest/Unhappiest Countries

Insight 10: Top 10 Happiest/Unhappiest countries world based. The top 10 happiest ranked countries are all European except Puerto Rico and Oman. Top 10 unhappiest countries are either African or Asian. Lowest ranking equals happiest.

Litteraturliste:

Figur 16. Wikipedia. “Flag of Puerto Rico”. January 2009.
https://en.wikipedia.org/wiki/Flag_of_Puerto_Rico#/media/File:Flag_of_Puerto_Rico.svg

Figur 16. Wikipedia. "Flag of Afghanistan". August 2021.

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